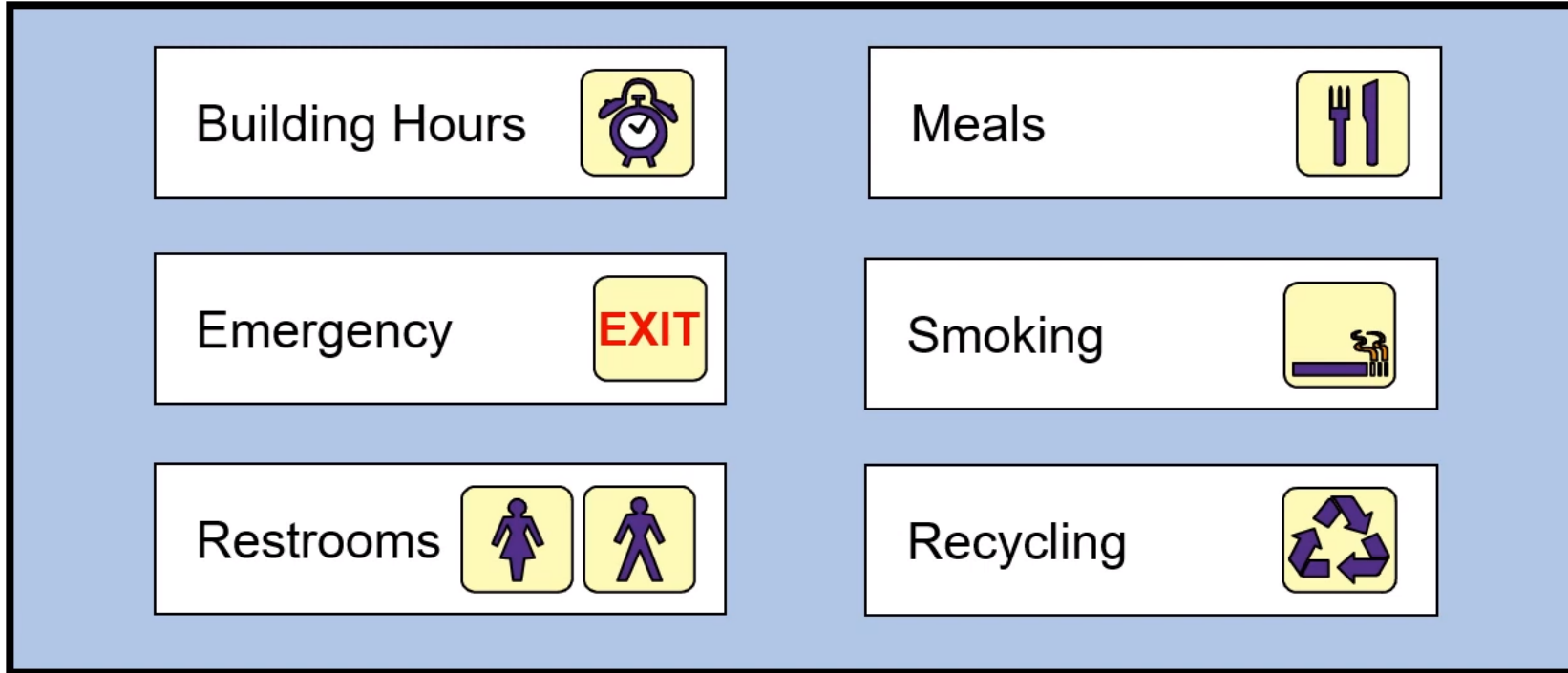


# IC Compiler II

## Block-level Implementation

2019.12-SP4

# Facilities



Please turn off or silence your cell phone

## IC Compiler II

- **This workshop will cover the entire block-level implementation flow:**
  - Design & Timing Setup
  - Floorplanning (introduction)
  - Placement and Optimization
  - Clock Tree Synthesis and Optimization
  - Routing and Post-Route Optimization, including InDesign PrimeTime and StarRC
  - Top-level Implementation, Signoff DRC, fill and ECO

# Workshop Target Audience

**ASIC, back-end, or layout designers who will be using *IC Compiler II* to perform placement, CTS and routing on block-level designs**



# Workshop Prerequisite Knowledge

- **Prior knowledge of IC Compiler or IC Compiler II is not needed**
- **Knowledge of general standard cell-based placement, CTS and routing concepts and terms is helpful**
- **An understanding of basic digital ASIC design concepts is assumed, including:**
  - Combinational and sequential logic functionality
  - Setup and hold timing

# Introductions

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- **Name**
- **Company**
- **Job Responsibilities**
- **Relevant Experience**
- **Main Goal(s) and Expectations for this Course**

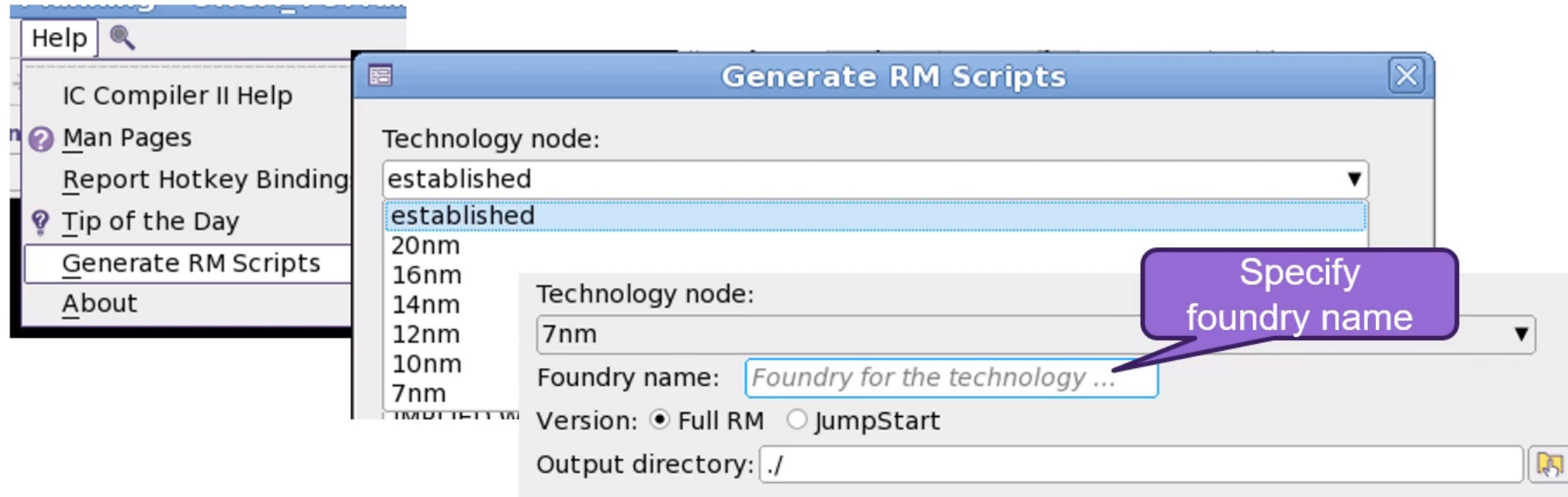
# IC Compiler II

- **Many things that you are used to doing in IC Compiler are no longer valid in IC Compiler II**
- **On the other hand, many concepts / use models are carried over to IC Compiler II**
  - Similar commands can have subtle or big differences!
  - Always consult the man page
- **To maximize the capabilities and performance of IC Compiler II, we recommend that you start with new ICC II Reference Methodology (RM) scripts**



# How To Access RM

- RM is built right into ICC II – Use the GUI or the Command Line



```
icc2_shell> generate_rm -rm_script JumpStart_2.0 -output ./ \
               -node <N> -foundry <F>
```

File `./ICC2-RM_P-2019.12-SP4.tar.gz` is generated. Please unpack the file to get IC Compiler II Reference Methodology scripts.



# IC Compiler II GUI

The screenshot displays the IC Compiler II GUI in the Design Planning mode. The main window shows a layout view with various components like M\_3, M\_2, M\_1, M\_2, M\_1, M\_8, M\_7, and various RAM blocks (e.g., \_0\_1, \_0\_2, \_0\_3, \_0\_4, \_1\_1, \_1\_2, \_1\_3, \_2\_1, \_2\_2, \_2\_3, \_2\_4, \_3\_1, \_3\_2, \_3\_3). The interface includes a menu bar (File, Task, Edit, Create, View, Select, Highlight, Window, Help), a toolbar with icons for input mode, rectangle intersect, partial selection, and replacement. A search bar is located at the top right. The 'Menu and Task Actions' panel is open, showing 'Placement.Placement' under 'Shell Commands (1)' and 'Application Options (15)'. The 'Visibility and Selection control' panel is also open, showing a list of objects with checkboxes for visibility and selection. The 'Layout view window' is highlighted, showing the main design area. The 'Menus Commands' panel is also highlighted, showing the menu bar. The 'Command Search' panel is highlighted, showing the search results for 'place\_opt'.

**Menus Commands**

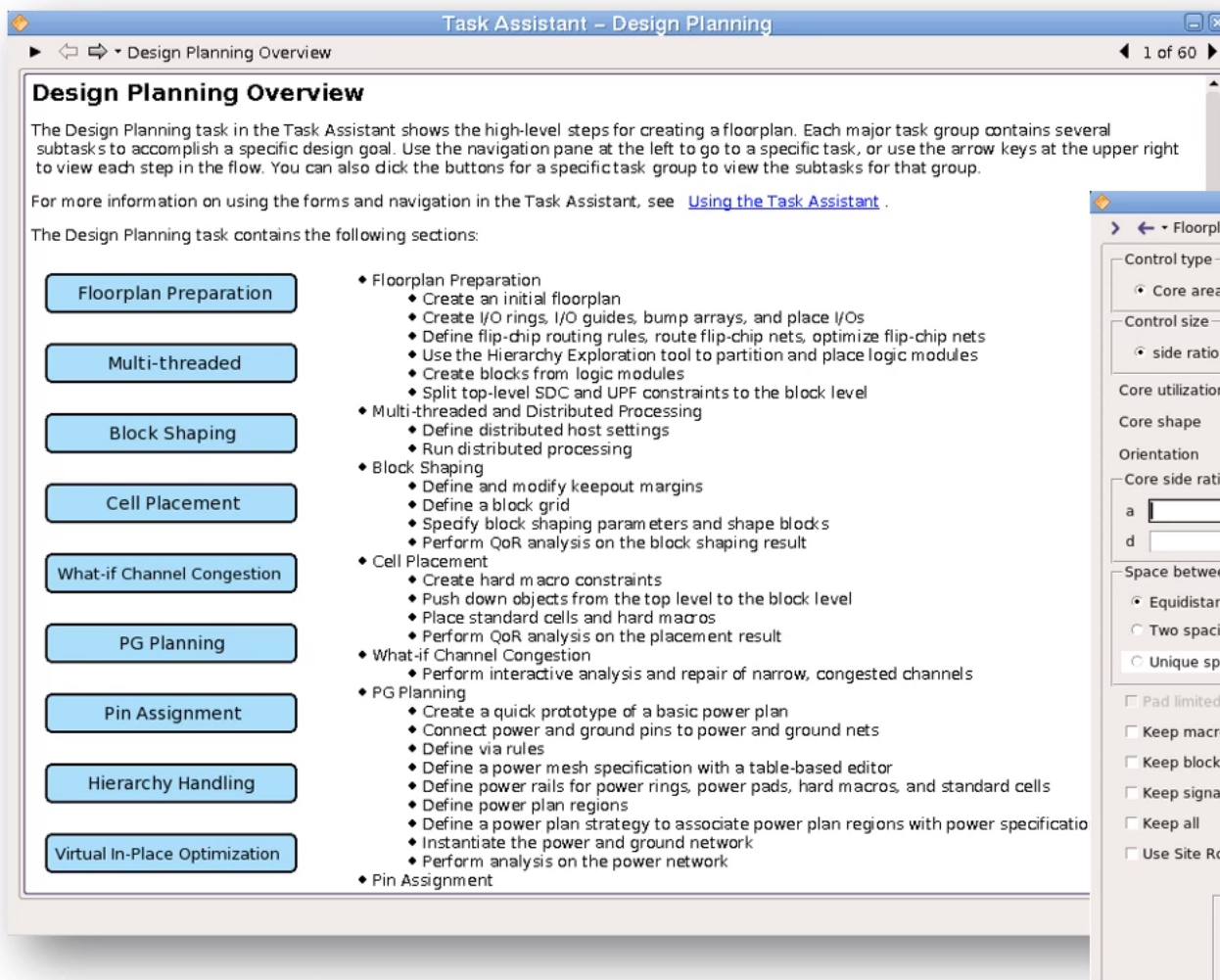
**Layout view window**

**Command Search**  
Example: List menus, commands and options related to "place\_opt"  
You can also run the commands from right here or pull up a dialog for setting application options.

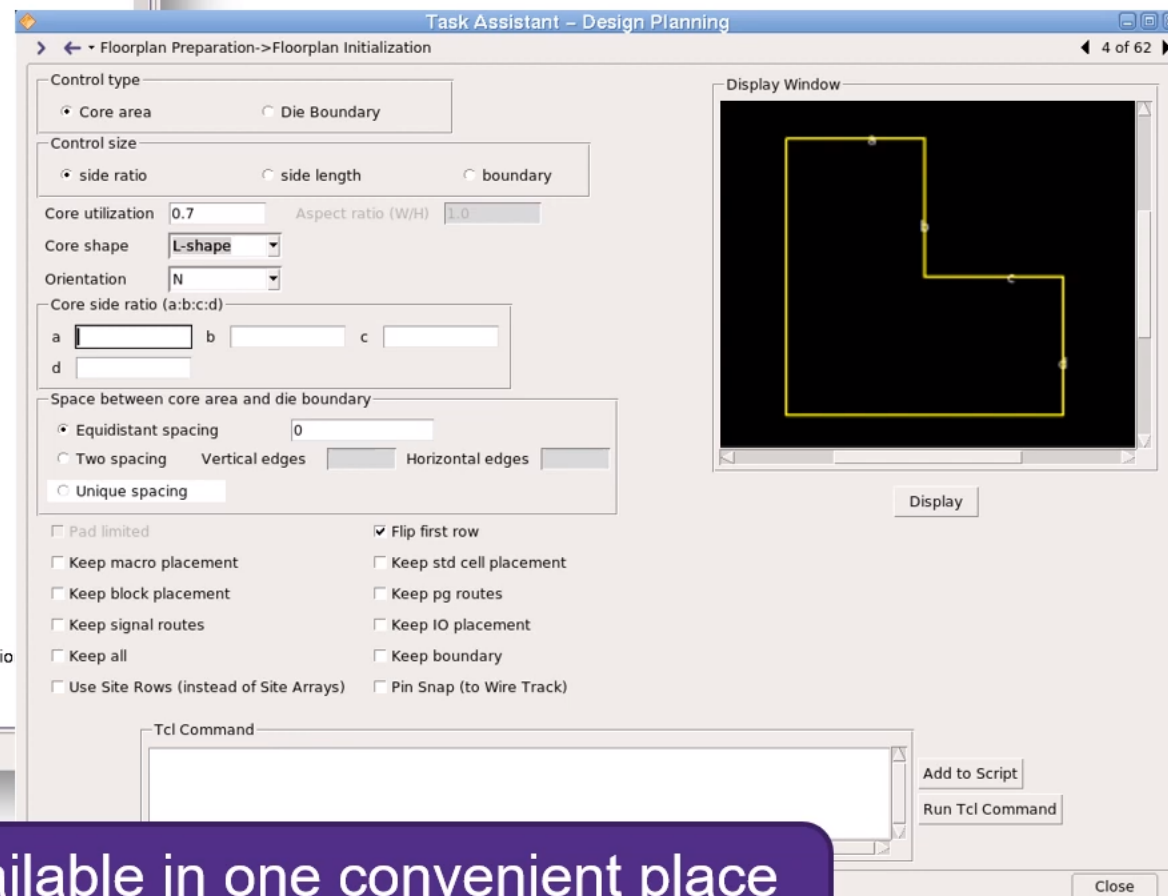
**Visibility and Selection control**

IC Compiler II - Design Planning - ORCA\_TOP.ndm:ORCA\_TOP/place\_opt.design [write]  
File Task Edit Create View Select Highlight Window Help  
Input mode: Smart (selected), Line  
Rectangle Intersect: Enable  
Partial Selection: Yes  
Replace: Replace  
Menu and Task Actions (1)  
Placement.Placement  
Shell Commands (1)  
place\_opt  
Application Options (15)  
place\_opt.congestion.effort  
place\_opt.final\_place.effort  
place\_opt.final\_place.legalize  
place\_opt.flow.clock\_aware\_placement  
place\_opt.flow.do\_path\_opt  
place\_opt.flow.do\_spg  
place\_opt.flow.optimize\_icgs  
place\_opt.flow.optimize\_icgs\_critical\_range  
place\_opt.flow.optimize\_layers  
place\_opt.flow.optimize\_layers\_critical\_range  
place\_opt.flow.optimize\_ndr  
place\_opt.flow.trial\_clock\_tree  
place\_opt.initial\_drc.global\_route\_based  
place\_opt.initial\_place.effort  
place\_opt.initial\_place.two\_pass  
64x32.frame  
\_FIFO\_RAM\_0  
64x32.frame  
Blockage  
Bound  
Cell  
Core Area  
Edit Group  
Fill Cell  
Guide  
Labels  
PG Region  
Pin  
Port  
Route  
Routing Corridor  
RP Blockage  
RP Group  
Site Array  
Site Row  
Terminal  
Text  
Track  
Via Region  
Voltage Area  
Preset: Default  
Click objects or drag a box to select (Hold Ctrl to add, Shift to remove)  
No Grid XY 807.731, 937.284 et: BUF\_net\_33959

# Task Assistant



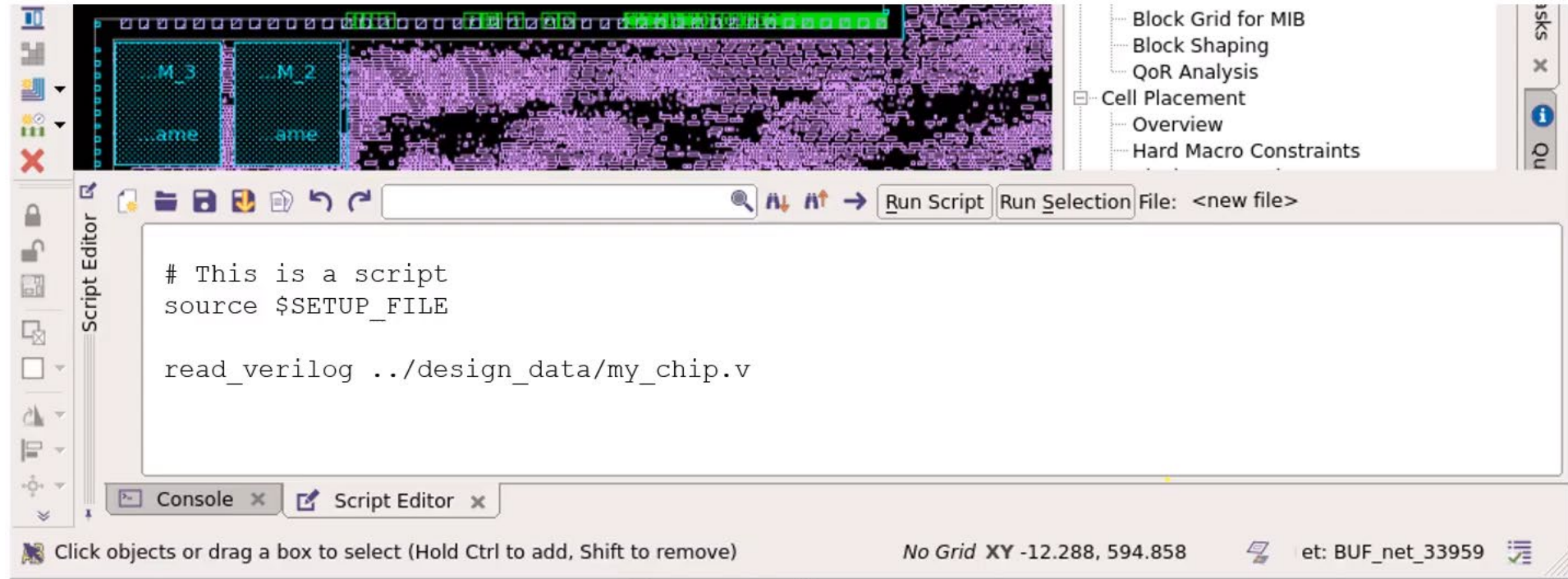
F4 launches the Task Assistant



Settings and commands available in one convenient place based on the recommended flow for the task at hand



# Built-In Script Editor



- Create scripts interactively
- Execute entire script or a selection
- Save script to a file

# Agenda

**DAY**

**1**

**i**

**Introduction and GUI Lab**



**1**

**Objects, Blocks and App Options**

**2**

**Floorplanning**



**3**

**Placement & Optimization**



# Agenda

**DAY**

**2**

**4** NDM Cell Libraries

**5** Design Setup



**6** Timing Setup



**7** Running CTS



# Agenda

**DAY**

**3**

**8**

**Setting Up CTS**



**9**

**Top Level Implementation**

**10**

**Routing & Optimization**

**11**

**Signoff**



**CS**

**Customer Support**